



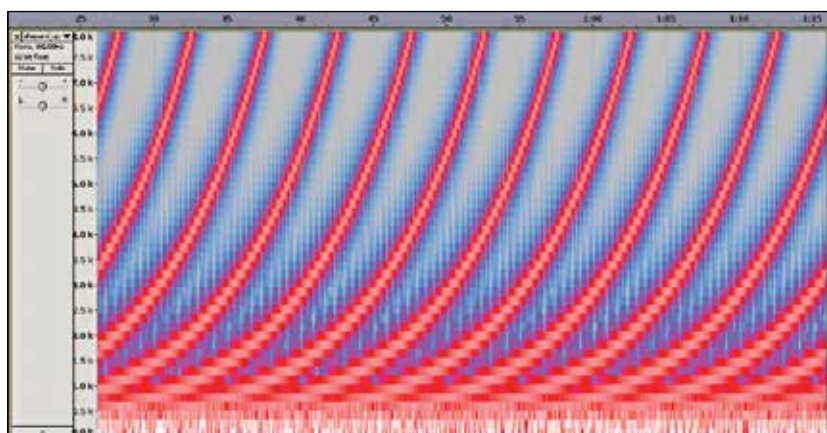
Maggots to save the world

Tasty waste-gobbling maggots that eat anything and turn them to organic compounds are the superheroes we need. <http://dgit.in/BioMaggot>



The Xbox One X is here

Microsoft finally kicks off pre-orders for their new Xbox One X console with a limited 'Project Scorpio' edition. <http://dgit.in/XboxOneX>



Shepard Tones downpitch, spectrum, linear-scaled

louder than the original sound, the discrete sounds are interpreted as one single sound. This effect is put to use to create the illusion of a stereo width starting from just a mono source.

For instance, if you have a mono source like guitars, it's a helpful trick to duplicate the original piece, pan one to the extreme right and the other to the left. Then delay either sample by about 10ms to 35ms (results may vary depending on application being used and particular sample). This makes the listener's mind perceive a lot of space and depth, and leaves the centre of the soundstage open for other instruments. This is also useful to make sure that a mono signal is not masked by other instruments at the centre.

FLETCHER-MUNSON EFFECT

Our ear is more sensitive to midrange sound, and as a result, relative levels of different instruments at different

frequencies changes depending on the volume you're listening to. This phenomenon is also referred to as the nonlinear response of the human ear. And it can be used to create the illusion of high volume even at low volume levels. This is the reason why professional sound mixers do so at low volumes (also to avoid ear fatigue).

By reducing the midrange or by boosting the high and low frequencies in a track, an impression of loudness at lower volumes can be created. For instance, just listen to Nirvana's Nevermind for a classic, thundering example of scooped mids dynamics. The same effect, when reversed, is useful in highlighting mid-range sounds – e.g. when you want to highlight the lead vocals versus backing vocals.

Interesting fact: The snare drum effect that you can hear in Johnny Cash's *Walk the Line* was created by inserting a piece of paper between the strings in his guitar.

TRITONE PARADOX

The tritone paradox, or the scale illusion, is a special case of Shepard's Paradox. In this, a pair of Shepard tones are separated by an octave, which might then be heard as ascending or descending.

In fact, there's a very interesting way to experience this paradox. Go to <http://dgit.in/TritonePrdx> and before you listen to the audio, make sure you've adjusted your headphones for equal loudness on each ear. Once you're done, play the audio and try to

understand what you're experiencing. Generally, you'll need to identify on which ear you're experiencing the higher and lower tone respectively. Now, turn the headphone around and listen again. Chances are that you'll be hearing the high and low tones in the same ear as before. This can be described as the brain assigning the higher and lower tones to different ears, with results differing between right handed and left handed people. Additionally, since both the tones are Shepard tones, they are heard to be continuous rising or falling without actually getting anywhere.

MCGURK EFFECT

The phrase 'seeing is believing' is one of the best descriptions for the McGurk effect. This effect refers to the combination of the audio of one phonetic syllable with the visual pronunciation of another leading to the listener hearing a third syllable. This is another example of the brain's conditioning that affects our experience of sound from beyond the auditory senses.



McGurk effect is nothing but your mind lip reading for you

For instance, if the audio of someone saying 'Ba' is combined with the visual of lips pronouncing 'ga' you're very likely to hear the syllable 'da'. While there are no practical benefits, hence applications, of this effect, it is pretty interesting to observe it for improving our understanding of how the brain processes inputs collectively from different senses. 

THE PHANTOM OF THE MELODIES

Sometimes music consists of patterns repeating at very high speed, with little or no difference between subsequent notes. When played fast enough, the brain identifies the subtly different notes and strings them together to form a melody. This same melody disappears when the sound is played slowly.